

TokTickIT — API Specification (Lab2)

1. Conventions

- **Base URL:** /api
- **Content-Type:** application/json for all requests/responses except file upload (multipart/form-data) and file download (binary passthrough with original Content-Type).
- **Requester context:** every Requester-scoped endpoint requires header X-Dev-Requester-Id: <requesterId> . This is a Lab 2 testing stand-in (BR-05) — never a client-supplied requesterId in the URL/query is trusted for ownership.
- **Errors:** a consistent envelope on all 4xx/5xx responses:

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Description is below the minimum length of 20 characters.",
    "fieldErrors": [
      { "field": "description", "message": "Must be at least 20 characters." }
    ]
  }
}
```

fieldErrors is present only for 400 validation failures; other error types omit it.

2. Reference Data Endpoints

GET /api/categories

Lists active Categories.

- **Auth:** none
- **Response 200:**

```
{ "data": [ { "id": 2, "name": "Hardware" }, { "id": 5, "name": "Software" } ] }
```

GET /api/related-systems

Lists active Related Systems.

- **Auth:** none
- **Response 200:**

```
{ "data": [ { "id": 7, "name": "Email" }, { "id": 3, "name": "VPN" } ] }
```

GET /api/priorities

Lists Priorities, ordered by `sortOrder` ascending.

- **Auth:** none
- **Response 200:**

```
{ "data": [  
  { "id": 1, "name": "Low", "sortOrder": 1 },  
  { "id": 2, "name": "Medium", "sortOrder": 2 },  
  { "id": 3, "name": "High", "sortOrder": 3 }  
]
```

GET /api/statuses

Lists Statuses.

- **Auth:** none
- **Response 200:**

```
{ "data": [  
  { "id": 1, "name": "New", "isDefault": true },  
  { "id": 2, "name": "Open", "isDefault": false },  
  { "id": 3, "name": "In Progress", "isDefault": false },  
  { "id": 4, "name": "Pending", "isDefault": false }  
]
```

Only the `isDefault` row is reachable as a Ticket's `currentStatusId` in Lab 2, since no

GET /api/dev-requesters

Lists active Development Requesters for the Selection screen (FR-01, BR-06).

- **Auth:** none
- **Response 200:**

```
{ "data": [ { "id": 12, "name": "Alex Rivera" } ] }
```

- **Response 200 (empty):** { "data": [] } — UI shows the empty-selector state.
- **Response 500:** generic safe error envelope.

3. Ticket Endpoints

POST /api/tickets

Creates a Ticket owned by the current Requester (FR-04).

- **Auth:** X-Dev-Requester-Id required.
- **Request body:**

```
{  
  "categoryId": 2,  
  "relatedSystemId": 7,  
  "summary": "Laptop battery drains quickly",  
  "description": "Battery drains fast even when idle, started after last update.",  
  "requestedPriorityId": 2  
}
```

- **Validation (BR-16–BR-18):**

Field	Rule
summary	required, trimmed, 5–150 chars
description	required, trimmed, 20–2000 chars
categoryId	required, must reference an active Category
relatedSystemId	required, must reference an active Related System
requestedPriorityId	required, must reference an existing Priority

- **Response 201:**

```

{
  "data": {
    "ticketNumber": "TKT-2026-000042",
    "requesterId": 1,
    "summary": "Laptop battery drains quickly",
    "description": "Battery drains fast even when idle, started after last update.",
    "categoryId": 2,
    "relatedSystemId": 7,
    "requestedPriorityId": 2,
    "currentStatusId": 1,
    "createdAt": "2026-08-22T09:14:00Z",
    "updatedAt": "2026-08-22T09:14:00Z"
  }
}

```

- **Response 400:** field-level validation errors (BR-16–BR-18), e.g. inactive/unknown categoryId , unknown requestedPriorityId .
- **Response 500:** no Ticket persisted (BR-20).

GET /api/tickets

Paginated, searchable, filterable, sortable list of the current Requester's own Tickets (FR-06–FR-10).

- **Auth:** scoped to X-Dev-Requester-Id .
- **Query parameters:**

Param	Type	Notes
search	string	matches Ticket Number / Summary, case-insensitive partial (BR-12)
category	int	Category id filter
requestedPriorityId	int	Priority id filter
currentStatusId	int	Status id filter (only the default New id is reachable in Lab 2)
sortBy	string	createdAt ticketNumber summary requestedPriorityId currentStatusId updatedAt
sortDir	string	asc desc , default per BR-14

Param	Type	Notes
page	int	default 1; invalid falls back to default (BR-15)
pageSize	int	default 10, capped 50; invalid falls back to default (BR-15)

Filters combine with AND logic and with any active search term (BR-13). Default sort: Created Date descending, Ticket Number descending as tiebreaker (BR-14).

- **Response 200:**

```
{
  "data": [
    {
      "ticketNumber": "TKT-2026-000042",
      "requesterId": 1,
      "summary": "Laptop battery drains quickly",
      "categoryId": 2,
      "requestedPriorityId": 2,
      "currentStatusId": 1,
      "createdAt": "2026-08-22T09:14:00Z",
      "updatedAt": "2026-08-22T09:14:00Z"
    }
  ],
  "page": 1,
  "pageSize": 10,
  "totalItems": 1,
  "totalPages": 1
}
```

GET /api/tickets/:ticketNumber

Full read-only detail for one owned Ticket (FR-11).

- **Auth:** rejects if not owner — returns 404, never 403 (BR-11).
- **Response 200:**

```

{
  "data": {
    "ticketNumber": "TKT-2026-000042",
    "summary": "Laptop battery drains quickly",
    "description": "Battery drains fast even when idle, started after last update.",
    "categoryId": 2,
    "relatedSystemId": 7,
    "requestedPriorityId": 2,
    "currentStatusId": 1,
    "createdAt": "2026-08-22T09:14:00Z",
    "updatedAt": "2026-08-22T09:14:00Z",
    "attachments": [
      {
        "attachmentId": 101,
        "originalFileName": "screenshot.png",
        "status": "ACTIVE",
        "uploadedAt": "2026-08-22T09:15:00Z"
      }
    ]
  }
}

```

- **Response 404:** Ticket not found, or exists but not owned by current Requester.

4. Attachment Endpoints

POST /api/tickets/:ticketNumber/attachments

Uploads an Attachment to an owned Ticket (FR-05, FR-12).

- **Auth:** rejects if not owner (404).
- **Request:** multipart/form-data , field file .
- **Validation:**

Rule	Source
Type in .jpg / .jpeg / .png / .webp / .pdf , checked by extension and MIME	BR-22
Max size 5 MB	BR-23

Rule	Source
Ticket must have fewer than 5 active Attachments	BR-24

- **Response 201:**

```
{
  "data": {
    "attachmentId": 102,
    "originalFileName": "invoice.pdf",
    "status": "ACTIVE",
    "uploadedAt": "2026-08-22T09:20:00Z"
  }
}
```

- **Response 404:** Ticket not found / not owned.
- **Response 413:** exceeds 5 MB.
- **Response 415:** type not in the allowed list.
- **Response 422:** would exceed the 5 active-Attachment limit.

GET /api/tickets/:ticketNumber/attachments

Lists Attachment metadata (active + removed) for an owned Ticket.

- **Auth:** rejects if not owner (404).
- **Response 200:**

```

{
  "data": [
    {
      "attachmentId": 101,
      "originalFileName": "screenshot.png",
      "status": "ACTIVE",
      "uploadedAt": "2026-08-22T09:15:00Z"
    },
    {
      "attachmentId": 99,
      "originalFileName": "old-log.pdf",
      "status": "REMOVED",
      "uploadedAt": "2026-08-20T10:00:00Z",
      "removedAt": "2026-08-21T08:00:00Z",
      "removalReason": "Duplicate of screenshot.png"
    }
  ]
}

```

GET /api/attachments/:attachmentId/download

Downloads an active Attachment (FR-13).

- **Auth:** rejects if not owner or not active (404 — BR-27 forbids downloading REMOVED Attachments).
- **Response 200:** binary file stream,
Content-Disposition: attachment; filename="<originalFileName>" .
- **Response 404:** not owned, not found, or status is REMOVED .

PATCH /api/attachments/:attachmentId/remove

Soft-removes an active Attachment (FR-14).

- **Auth:** rejects if not owner (404).
- **Request body:**

```
{ "reason": "Wrong file, replaced by newer screenshot." }
```
- **Validation:** reason required, non-empty (BR-26).
- **Response 200:**


```

{
  "data": {
    "attachmentId": 101,
    "status": "REMOVED",
    "removedAt": "2026-08-22T10:00:00Z",
    "removalReason": "Removed"
  }
}

```

- **Response 400:** missing/empty reason .
- **Response 404:** not owned / not found.
- **Response 409:** already REMOVED .

8. Status Code Reference

Status	Used For
200	Successful retrieval (list, detail, attachment/comment/action/log metadata, download, remove)
201	Ticket created; Attachment uploaded
400	Validation failure
404	Ticket/Attachment/Comment/Action/Log not found, or not owned by the current Requester (BR-11)
409	Soft-remove attempted on an already-removed Attachment
413	Attachment exceeds 5 MB
415	Attachment type not in the allowed list
422	Upload would exceed the 5 active-Attachment limit
500	Unexpected server error (generic, safe message only — no stack traces to client)